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PetroML Dashboard

Business Value & AI Methodology

Machine-Learning Prediction of Porosity & Permeability from Petrographic Point-Counting Data

Based on: Sadrikhanloo, Busch & Hilgers (2026)

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1. Executive Summary

This document describes the business value proposition and AI methodology implemented in the PetroML Streamlit dashboard. The dashboard operationalises the machine-learning workflow published by Sadrikhanloo, Busch & Hilgers (2026) in Energy Geoscience, which predicts porosity and permeability from petrographic point-counting data using Random Forest and Support Vector Regression models with SHAP-based explainability.

The core innovation is the use of widely available but underutilised petrographic datasets — classical point-counting analyses of detrital and authigenic mineral phases, grain coating coverages, and granulometry — as input features for supervised regression. This approach bridges a gap between geophysical well-log-based ML (which cannot capture diagenetic controls) and laboratory core analysis (which is expensive and spatially limited).

The dashboard provides six operational modules: data exploration, porosity prediction (Random Forest), permeability prediction (SVR), SHAP-driven explainability of reservoir quality controls, single-sample real-time inference, and a business case visualisation with cost-comparison analytics.

2. Problem Statement & Industry Context

Reservoir quality assessment in siliciclastic formations has historically depended on three data sources: routine core analysis (RCAL) for porosity and permeability, petrographic thin-section analysis for mineralogy and diagenetic interpretation, and wireline well logs for continuous downhole properties. Each has limitations:

Core analysis is the gold standard but is expensive ($50–150/ft acquisition, $500–2,000/plug for RCAL), spatially discontinuous, and sometimes impossible due to core loss or slim economic margins in geothermal, CCS, or brownfield extension projects.

Wireline logs provide continuous coverage but cannot differentiate critical diagenetic factors — for example, they cannot distinguish radial from tangential illite coatings, which have opposite effects on quartz cementation and chemical compaction.

Petrographic data from point-counting analyses is widely available in legacy archives but is typically used only for qualitative paragenetic interpretation, not quantitative property prediction.

The paper demonstrates that petrographic point-counting data can be used as input features for ML models that predict porosity and permeability with acceptable accuracy, while simultaneously revealing diagenetic reservoir quality controls through SHAP explainability. The dashboard operationalises this workflow for real-time inference and business decision support.

3. Data Architecture

3.1 Source Dataset

The original study used 157 samples from six wells across four regions in central Europe, covering two major reservoir lithologies: the Permian Rotliegendes (Northern Germany, Southern Permian Basin) and the Triassic Buntsandstein (three regions in the Upper Rhine Graben area, Southern Germany). The dataset spans a porosity range of 0–20% and a permeability range of 0.0001–780 mD (nearly seven orders of magnitude).

3.2 Feature Engineering

The dataset contains 62 petrographic features. The dashboard implements a representative subset of 18 features organised into three categories:

Category	Feature	Geological Significance
Detrital Composition	Quartz, K-Feldspar, Plagioclase, MRF undiff, Ductile RF	Framework mineralogy controls compaction behaviour and cement precipitation sites
Authigenic Phases	Auth. TiOx, Calcite Cement, Quartz Cement, K-Feldspar Cement, Dolomite	Pore-filling cements reduce porosity; TiOx records K-feldspar dissolution (secondary porosity)
Porosity Types	Intergranular Porosity, Optical Porosity, IG Por. K-Fsp	Optically determined porosity from point-counting; underestimates microporosity vs. plug measurements
Clay Minerals & Coatings	GTI Coating, GTG Coating, Pore-filling Illite, Pore-lining Illite	Coating coverage and clay texture control quartz cementation inhibition vs. chemical compaction enhancement
Texture	Grain Size (mm)	Controls pore throat diameter and permeability

3.3 Synthetic Data Generation

Because the original supplementary data was not bundled with the dashboard, a geologically-informed synthetic data generator produces 157 samples with realistic feature–target relationships. The generator encodes the diagenetic controls identified by the paper's SHAP analysis:

Porosity-positive controls: pore-filling (radial) illite (+0.35 per %), authigenic TiOx (+1.5 per %), MRF undiff (+0.22 per %), intragranular K-feldspar porosity (+0.8 per %), K-feldspar cement (+0.20 per %), grain size (+1.0 per mm).

Porosity-negative controls: calcite cement (−0.6 per %), quartz cement (−0.4 per %), pore-lining illite above 3% threshold (−0.20 per excess %), ductile RF (−0.18 per %), quartz grains above 55% (−0.12 per excess %).

Nonlinear coating effects: GTI coating below 40% is porosity-positive (radial coatings inhibit quartz cementation); above 40% it becomes porosity-negative (tangential coatings at grain contacts enhance chemical compaction). GTG coating shows a similar inflection at 60%.

Critically, intergranular and optical porosity are computed independently from different feature subsets with substantial noise ($\sigma = 2.2$), preventing the Random Forest from trivially short-circuiting through a single proxy variable. This ensures distributed feature importance across 6+ features, matching the paper's multi-factor SHAP results.

4. AI Methodology

4.1 Porosity Prediction — Random Forest

The porosity model uses a Random Forest regressor (Breiman, 2001), an ensemble method that constructs multiple decision trees and averages their outputs. RF was selected because it captures nonlinear relationships and feature interactions in high-dimensional datasets with limited samples.

4.1.1 Hyperparameters

Parameter	Paper Value	Dashboard Value
n_estimators	Bayesian-optimised	200
max_depth	Bayesian-optimised	12
min_samples_split	Bayesian-optimised	4
min_samples_leaf	Bayesian-optimised	2
Cross-validation	5-fold	5-fold
Train/Test split	80/20 shuffled	80/20 shuffled

4.1.2 Performance Metrics (Paper)

The paper reports test-set metrics of $R^2 = 0.92$, MAE = 1.25%, and RMSE = 1.56%. Cross-validation mean $R^2 = 0.86 \pm 0.06$. The dashboard's synthetic data model achieves $R^2 \approx 0.74$–0.88 depending on noise realisation, which is lower due to the deliberately increased noise in the synthetic intergranular porosity feature (designed to distribute importance rather than concentrate it).

4.2 Permeability Prediction — Support Vector Regression

Permeability prediction uses Support Vector Regression with a linear kernel. SVR was chosen because it effectively handles high-dimensional small-sample problems where the number of features may exceed the number of samples. The linear kernel consistently outperformed radial basis function kernels in both cross-validation and test evaluation.

4.2.1 Data Preprocessing

Log-transformation: Permeability values spanning nearly seven orders of magnitude (0.0001–780 mD) are $\log_{10}$-transformed before training to stabilise variance and improve model interpretability. All error metrics are reported in both real-scale and log-space.

RobustScaler: Both features and log-transformed permeability targets are scaled using sklearn's RobustScaler, which uses median and interquartile range rather than mean and standard deviation, ensuring robustness against outliers — critical for datasets with extreme permeability ranges.

4.2.2 Performance Metrics (Paper)

Real-scale: $R^2 = 0.81$, MAE = 29.4 mD, RMSE = 68.4 mD, NRMSE = 8.76%. Log-transformed: $R^2 = 0.83$, MAE = 0.21, RMSE = 0.24 (orders of magnitude). Predictions generally fall within 0.21–0.24 orders of magnitude of measured values — within one order of magnitude, which is appropriate for practical reservoir characterisation.

4.3 SHAP Explainability

SHapley Additive exPlanations (SHAP) values quantify the contribution of each feature to individual predictions, providing model-agnostic interpretability. The dashboard implements two SHAP approaches:

TreeExplainer (porosity model): Exact SHAP values computed efficiently for tree-based models using the algorithm of Lundberg & Lee (2017). Applied to the full training set for robust importance estimates.

KernelExplainer (permeability model): Model-agnostic SHAP computation for the SVR, using a background sample of 50 training instances and evaluated on a 15–30 sample test subset for computational efficiency.

4.3.1 Key Geological Insights from SHAP

The SHAP analysis reveals a critical sign-flip between porosity and permeability models for pore-filling (radial) illite:

Porosity model (positive): Radial illite coatings inhibit syntaxial quartz overgrowth cementation, preserving macropore volume. Higher pore-filling illite content increases predicted porosity.

Permeability model (negative): The same radial and meshwork illite textures reduce effective pore throat radii, impeding fluid flow. Higher pore-filling illite content decreases predicted permeability (Neasham, 1977).

This dual behaviour — porosity-preserving but permeability-reducing — is a well-documented diagenetic phenomenon that the ML models correctly capture without being explicitly programmed to do so. The dashboard presents separate geological interpretation panels for each model tab to avoid conflating these opposite effects.

Additional SHAP-derived controls include: authigenic TiOx as a provenance-specific proxy for K-feldspar dissolution in the Buntsandstein; the GTI coating threshold at ~40% separating radial (porosity-preserving) from tangential (compaction-enhancing) illite textures; and the negative effect of calcite cement (exclusively Rotliegendes) on both porosity and permeability.

5. Dashboard Architecture

The PetroML dashboard is built as a single-file Streamlit application (Python) with six navigable sections:

Module	Description & Key Components
Overview & Data	Interactive porosity–permeability cross-plot (Plotly, log-scale), well-coded symbology, dataset preview with well filtering. Metric cards showing sample count, well/region coverage, feature count, and stratigraphic groups.
Porosity Model (RF)	Measured vs. predicted scatter plot with 1:1 reference line, test-set R^2/MAE/RMSE cards, 5-fold CV R^2 with standard deviation, and horizontal bar chart of Gini feature importance.
Permeability Model (SVR)	Log-transformed measured vs. predicted scatter with 1:1, 1:10, and 10:1 reference lines. Real-scale and log-transformed permeability histograms. Test metrics in log-space.
SHAP Explainability	Tabbed interface (Porosity RF / Permeability SVR). Each tab: mean |SHAP| bar chart, interactive dependence plot selector with secondary-feature colour coding. Model-specific geological interpretation panels.
Single-Sample Predictor	18 slider inputs for point-counting data. Real-time porosity and permeability prediction with gauge visualisations. Permeability gauge shows real mD tick labels on log axis. Reservoir quality classification (Very Tight to Good–Excellent).
Business Case	Six value-driver cards, stacked-bar cost comparison (Conventional vs. ML-Augmented vs. Cuttings-Only), and paper citation.

5.1 Technology Stack

Python 3.10+, Streamlit ≥1.30, scikit-learn ≥1.3 (RandomForestRegressor, SVR, RobustScaler, cross_val_score), SHAP ≥0.43 (TreeExplainer, KernelExplainer), Plotly ≥5.18 (interactive charts, gauge indicators), Pandas, NumPy. Designed for deployment on Streamlit Community Cloud or Azure App Service (aligned with Digital Oil's Azure-based infrastructure).

6. Business Value Proposition

6.1 Cuttings-Based Reservoir Evaluation

The most transformative application is predicting porosity and permeability from drill cuttings — material that is continuously available during drilling but rarely used for quantitative property estimation. By preparing petrographic thin sections from cuttings (skipping areas outside individual fragments per Ölmez et al., 2025) and applying standardised point-counting, the trained model can provide near-real-time reservoir quality estimates during drilling operations. This extends subsurface characterisation to well sections where core was never taken, was lost, or was not economically justified.

6.2 Reduced Coring & Laboratory Costs

Core acquisition typically costs $50–150/ft, with routine core analysis adding $500–2,000 per plug sample. By training models on existing core data from a formation and extending predictions to uncored intervals via cuttings or legacy thin sections, operators can reduce coring programmes by an estimated 30–50% while maintaining equivalent subsurface insight. The dashboard’s cost comparison module illustrates potential savings of $100–190K per well when moving from full-core to ML-augmented or cuttings-only approaches.

6.3 Legacy Data Monetisation

Decades of petrographic point-counting datasets sit underutilised in operator and geological survey archives. Public databases in the Netherlands (NLOG) and Norway (DISKOS) already provide petrographic and core analysis data for numerous wells. This ML workflow converts these static, qualitative reports into predictive models that can forecast reservoir quality in new wells within the same play, extracting value from sunk data-acquisition costs.

6.4 SHAP-Driven Diagenetic Understanding

SHAP explainability goes beyond prediction accuracy — it provides a systematic, data-driven tool for identifying which mineral phases and textures control reservoir quality. For the studied dataset, SHAP analysis independently recovered all previously published reservoir quality controlling processes: enhanced chemical compaction along illite-coated grain contacts, radial illite coatings inhibiting quartz cementation, and radial illite reducing permeability despite preserving porosity. This approach may reveal previously unrecognised controls when applied to new datasets, supplementing traditional paragenetic analysis.

6.5 Energy Transition Applicability

Geothermal, CO_2 storage (CCS), and subsurface hydrogen storage projects often operate on slim economic margins that do not justify dedicated coring programmes. Models trained on petroleum-era core data from the same formations can de-risk these new-energy reservoirs using petrographic analysis of cuttings alone, substantially lowering exploration and appraisal costs.

6.6 Operational Integration

The workflow integrates with standard E&P data workflows — LAS/DLIS well logs, Excel-based petrophysical databases, and petrographic data management systems. Outputs (predicted porosity, permeability, reservoir quality classification) can feed directly into static earth models, well placement optimisation, and completion design. The single-sample predictor module enables geologists to run inference on individual point-count analyses without requiring data-science expertise.

7. Cuttings-Based Reservoir Characterisation Workflow

The most transformative near-term application of this ML framework is predicting reservoir properties from drill cuttings — rock fragments continuously produced during drilling and recovered at the shale shaker. Unlike cores, cuttings are available at 3–10 ft intervals across the entire drilled section at effectively zero marginal collection cost. This section describes the end-to-end workflow from cuttings recovery to ML inference.

7.1 Cuttings Collection & Preparation

Cuttings are collected at the shale shaker during drilling, washed to remove drilling fluid contamination, dried, and catalogued at regular depth intervals (typically 3–10 ft depending on ROP and mud system). For petrographic analysis, cuttings of suitable size and quality are selected, mounted in epoxy, and prepared as standard 30 μm thin sections. During point-counting, the petrographer systematically excludes areas outside individual rock fragments to avoid counting mounting medium or contamination (per the methodology of Ölmez et al., 2025, applied to flysch play cuttings in the Vienna Basin).

7.2 Direct Porosity Measurement from Cuttings

A critical advantage of the cuttings-based approach is that porosity can be directly measured on cuttings material, providing the single most important feature for permeability prediction. The paper identifies multiple established laboratory methods for cuttings porosity:

Archimedes' principle: Simple buoyancy-based measurement using helium or water saturation, applicable to cuttings fragments >5 mm.

Mercury injection capillary pressure (MICP): Provides both porosity and pore-throat size distribution from small rock chips; pore throat data directly correlates with permeability.

Micro-CT (μCT): Non-destructive 3D imaging at sub-micron resolution, yielding connected porosity, pore network topology, and pore-throat geometry (Hübner, 2014).

Nuclear Magnetic Resonance (NMR): Provides total porosity, bound vs. free fluid volumes, and pore size distribution from small cuttings samples (Miritchnik et al., 2004; Kesserwan et al., 2017).

Including measured cuttings porosity alongside petrographic point-count data significantly enhances permeability prediction, because the SVR model's strongest predictor is petrophysical porosity (SHAP values ranging from −0.6 to +0.6). Porosity from cuttings, while potentially affected by sample preparation, provides a sufficiently accurate proxy to substantially improve model performance versus petrography alone.

7.3 Data Continuity Advantage

Core material typically covers 10–15% of the drilled reservoir section, with acquisition costs of $50–150/ft and frequent core loss. Cuttings provide 85–95% coverage of the drilled section, constrained only by intervals where cuttings returns are poor (e.g., lost circulation zones, underbalanced sections). This near-continuous coverage enables:

Depth-continuous reservoir quality profiles: Rather than isolated porosity–permeability measurements at cored depths, the ML model produces continuous predicted profiles that can be directly correlated with wireline log responses.

Diagenetic facies mapping: Systematic point-counting of cuttings across the reservoir section reveals spatial variations in authigenic cement distribution, coating coverage, and dissolution porosity that may not be captured by cores at discrete intervals.

Bypassed pay identification: Predicted porosity and permeability profiles from cuttings may reveal reservoir-quality intervals that were not targeted for core acquisition, enabling identification of previously overlooked pay zones.

7.4 Calibration Requirements

The accuracy of cuttings-based predictions depends critically on the availability of core calibration data from the same or analogous formation. The model must be trained on core samples that capture the range of diagenetic alteration and compaction encountered in the target lithology. Without suitable calibration data, predictions may be inaccurate due to incorrect assessment of controlling factors or local variations in diagenetic overprint. The paper notes that local variability within formations — such as differences in clay mineral coating types (illite, chlorite, or kaolinite in the Rotliegendes) or lithofacies changes between marginal and basinal Buntsandstein facies — requires formation-specific calibration.

8. Geochemical Data Integration

The paper explicitly identifies geochemical and mineralogical datasets as candidates for integration: “further datasets, for example geochemical datasets (e.g., from XRF/pXRF/µXRF, EDX) or mineralogical datasets (e.g., XRD, mineral mapping), can be easily integrated and utilised to maximise the knowledge gain from subsurface samples.” This section details the technical and business value of geochemical feature augmentation.

8.1 Portable XRF (pXRF) as an Operator-Independent Feature Source

Portable X-ray fluorescence (pXRF) provides rapid, non-destructive elemental analysis of cuttings and core material at $15–40 per sample. Key elements relevant to the PetroML feature set include:

Titanium (Ti): The paper identifies authigenic TiOx as a strong positive predictor of porosity, linked to K-feldspar dissolution. A pXRF-derived Ti/Al ratio provides the same provenance signal without subjective mineral identification, at a fraction of the cost of SEM-EDS analysis.

Potassium (K): K content correlates with K-feldspar and illite abundance, both of which are key features in the ML model. K/Al ratios distinguish illite-rich from kaolinite-rich assemblages.

Calcium (Ca): Ca content directly proxies calcite cement volume, identified by SHAP as a strong negative predictor of porosity in the Rotliegendes. pXRF Ca measurements are faster and more objective than point-counting for quantifying carbonate cement.

Iron (Fe) and Manganese (Mn): Fe/Mn ratios distinguish different carbonate cement generations (ferroan vs. non-ferroan dolomite, siderite vs. ankerite), refining the diagenetic interpretation beyond what optical petrography alone can provide.

The fundamental advantage of pXRF over point-counting for these specific features is objectivity: pXRF measurements are not subject to inter-operator variability, which the paper identifies as a key limitation of petrographic datasets spanning multiple decades and providers.

8.2 X-Ray Diffraction (XRD) for Quantitative Mineralogy

XRD with Rietveld refinement provides quantitative bulk mineral phase fractions at $30–80 per sample. For the PetroML feature set, XRD can independently quantify:

Total illite content: distinguishing bulk illite from the point-count differentiation of pore-filling (radial/meshwork) vs. pore-lining (tangential) textures. XRD provides the total clay volume, while petrography provides the textural interpretation — the combination is more powerful than either alone.

Quartz vs. feldspar vs. lithic fractions: independent validation of the QFR classification used in the paper, without reliance on optical identification of altered or fine-grained phases.

Authigenic vs. detrital phase proportions: When combined with grain-size-specific XRD (clay-fraction analysis), the contribution of authigenic clays to bulk mineralogy can be separated from detrital clay matrix.

8.3 Micro-XRF and EDX for Spatial Geochemistry

Micro-XRF (µXRF) and energy-dispersive X-ray spectroscopy (EDX) provide spatially-resolved element maps on polished thin sections at $50–150 per sample. These methods can validate and augment the most texturally-dependent features in the model:

Grain coating composition at GTI/GTG contacts: µXRF mapping can chemically distinguish illite from chlorite grain coatings, which have different effects on quartz cementation inhibition and chemical compaction. This addresses a feature that point-counting captures positionally (coating coverage %) but cannot always resolve compositionally.

Cement stratigraphy: Spatially-resolved element maps reveal zoning in authigenic quartz, feldspar, and carbonate cements, providing information on diagenetic paragenesis that can be encoded as additional features for the ML model.

8.4 Extended Feature Matrix: Technical Architecture

The integration of geochemical data with point-counting creates a complementary feature matrix. Point-counting excels at features that geochemistry cannot replicate: coating coverage (GTI/GTG percentages), specific porosity types (intergranular vs. intragranular), and grain size/sorting. Geochemistry fills gaps in elemental composition, quantitative bulk mineralogy, and 3D pore-network characterisation. The combined feature set is both richer and more objective than either source alone.

From a model architecture perspective, the existing RF and SVR frameworks can accommodate additional geochemical features without modification — they simply expand the feature vector from 18 to 25–40 columns. SHAP analysis on the extended feature set would then reveal whether geochemical proxies provide additional explanatory power beyond the point-counting features, or whether they are redundant. The paper's observation that RFE showed no significant improvement beyond 45 features suggests that the model can accommodate a moderately expanded feature set without overfitting.

8.5 Cost–Coverage–Objectivity Trade-Offs

The geochemistry-augmented cuttings approach fundamentally reshapes the cost–coverage trade-off for reservoir characterisation. A conventional full-core programme with RCAL costs approximately $230–240K per well but covers only 10–15% of the reservoir section. A cuttings-based programme with point-counting, pXRF, and XRD covers 85–95% of the section at approximately $40K per well — roughly 17% of the conventional cost with 6× the coverage. Adding µCT for pore-network analysis raises the cost to ~$60K but provides 3D pore connectivity data that correlates directly with permeability.

Crucially, the geochemistry-augmented approach also reduces operator bias risk from "moderate" (subjective point-counting) to "low" (objective instrumental measurement), addressing the primary limitation identified in the paper for scaling this methodology beyond a single research group's dataset.

9. Limitations & Future Work

Operator bias: The study used data from only two petrographers at the same institution. Production deployment would need to account for inter-operator variability in point-counting categories and identification of mineral phases. Geochemical augmentation (Section 8) mitigates but does not eliminate this issue, as textural features (coating coverage, porosity type) remain dependent on point-counting.

Dataset size: 157 samples from six wells (two stratigraphic units) limits generalisability. Well-wise splitting was not enforced due to insufficient per-well sample counts. Future applications should incorporate blind-well validation.

Depth-dependent trends: Shuffled data prevents the model from capturing fining- or coarsening-upward sequences. Spatially-aware models could improve predictions for continuous well sections.

Cuttings fabric preservation: Cuttings may experience mechanical damage during drilling (bit grinding, transport in annulus) that disturbs original rock fabric. Coarser cuttings from PDC bits generally preserve fabric better than fine cuttings from roller-cone bits. The degree of fabric preservation affects the reliability of point-counting data from cuttings and must be assessed on a case-by-case basis.

Synthetic data limitations: The dashboard's synthetic data generator approximates but does not replicate the full 62-feature dataset or the specific mineralogical variability of the original formations. Production deployment should use actual supplementary data or operator datasets.

Scalability: SVR training time increases with sample count. For larger datasets (>1,000 samples), gradient boosting methods (XGBoost, LightGBM) or neural network approaches may be more appropriate.

10. Conclusion

The PetroML dashboard demonstrates a viable pathway from academic research to operational tool, translating the Sadrikhanloo, Busch & Hilgers (2026) workflow into a deployable, interactive application. The combination of Random Forest porosity prediction ($R^2 = 0.92$), SVR permeability prediction (log-space $R^2 = 0.83$), and SHAP-based geological explainability provides both predictive accuracy and interpretive value — a critical requirement for adoption by domain experts who need to understand why a model makes its predictions, not just what it predicts.

The extension to cuttings-based reservoir characterisation, augmented by geochemical data from pXRF, XRD, and μCT, transforms the cost–coverage equation: from $230K per well with 10–15% section coverage to $40–60K per well with 85–95% coverage and reduced operator bias. This approach is directly applicable to energy-transition scenarios (geothermal, CCS, subsurface H_2 storage) where slim economic margins do not justify conventional coring programmes.

By operationalising this combined petrographic–geochemical–ML approach through Digital Oil's Azure-based SaaS platform, the methodology becomes accessible to operators without in-house data science capabilities, while the SHAP-driven insights enhance reservoir understanding for all users.

Reference: Sadrikhanloo, S., Busch, B., Hilgers, C. (2026). Porosity and permeability prediction from petrographic point-counting data using machine learning: Applications to Rotliegendes and Buntsandstein reservoirs. Energy Geoscience 7, 100537. https://doi.org/10.1016/j.engeos.2026.100537